

Pre-feasibility study for Building Stone Project



2017



Table of Contents

1. EXECUTIVE SUMMARY	5
2. KEY HIGHLIGHT OF JORDAN	6
3. MARKET ANALYSIS	13
<i>Project description</i>	13
<i>Project Objectives</i>	14
<i>Limestone uses</i>	Error! Bookmark not defined.
<i>Target Segments</i>	16
<i>Market Size Analysis</i>	16
<i>Competitor analysis</i>	17
<i>Price analysis and pricing policy</i>	17
<i>Market Share</i>	17
<i>Estimated Revenues</i>	19
4. TECHNICAL ANALYSIS	20
<i>Human resorces required</i>	20
<i>Location analysis</i>	20
<i>Description of production process and steps of manufacturing</i>	21
<i>Technical requirement analysis</i>	22
1.1.1 <i>Land and construction works</i>	22
<i>Equipments and machinery</i>	22
<i>Furnitures</i>	23
<i>Requirments and legal procedures</i>	23
5. FINANCIAL ANALYSIS	25
<i>Assumptions</i>	26
<i>Capital Expenditures</i>	28
<i>Operating Expenses</i>	31
<i>General and Administrative Expenses</i>	32
<i>Marketing Expenses</i>	32
<i>Human Resources</i>	33
<i>Depreciations</i>	35
<i>Loan</i>	36
<i>Income Statement</i>	37
<i>Expected Cashflows Statement</i>	39
<i>Expected Balance Sheet</i>	41
<i>Feasibility Indicators</i>	43
<i>Sensitivity Analysis</i>	44

List of Tables

Table1 Project Main results.....	5
Table 2: Population Distribution according to Age Group:.....	7
Table 3 Income Tax and Exemptions.....	12
Table 15 General Assumptions.....	26
Table 16 Currency Exchange Rates.....	26
Table 17 Annual Growth Rates Assumptions	26
Table 18 Expenses Assumptions.....	26
Table 19 Income Tax Assumptions	27
Table20 : Weighted Average Cost of Capital.....	27
Table 22 Construction Work Capital Expenditure.....	28
Table 22 machinery and equipments Capital Expenditure.....	28
Table 22 Furniture and furnishings Capital Expenditure	29
Table: 25 Transporation and Vehicles Capex	29
Table : 26 Pre Operating Expenses.....	29
Table27 Working Capital	29
Table 28 Expenses Summary	30
Table 29 Sources of Funding.....	30
Table 30 Operating Expenses	31
Table 31 General and Administrative Expenses.....	32
Table 32 Marketing Expenses	32
Table : 29 Staff numbers expected during (2027-2018).....	33
Table 30 Expected monthly salary during (2027-2018).....	34
Table : 33 Expected Annual Salaries and Wges (2018-2027)	34
Table 35 Capex and Depreciation Expenses.....	35
Table 36 Depreciations and Additions on Construction Works	35
Table : 37 Loan Details	36
Table 38 Income Statement.....	37
Table 39 Expected Cashflows	39
Table 40 Expected Balance Sheet.....	41
Table 41 Free Net Cash flows Table	43
Table 42 Payback Period	43
Table 43 Financial Analysis Results	44
Table 44 Sensitivity Analysis	44

List of Figures

Figure 1 Population of Jordan.....	6
Figure2 Sectors Contribution to Jordan's Economy	8

1. Executive summary

Limestone is one of the most ore sourced in a large amount in Jordan, so establishing a quarry to extract limestone ores with high quality, will allow to it compete in national and international markets.

Based on proposed market share and the prices, the expected revenues will be about 3,228,000 JD and it will be a feasible project during the first year of the project life, and will be rising to 3,820,376 JD during to tenth year. The payback period is four year, the internal rate of return is 28.97%

The following table summarize the project main results:

Table1 Project Main results

Results	Value (JD-Years)
Total Investment Cost (Jordan Dinars)	1,799,039
Internal Rate of Return (IRR)	%28.97
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	2,608,617
Job Opportunities created by the Project	11
Total Revenues (For 10 Years)	33,866,239
Total Operating Expenses (For 10 years)	27,380,854
Total Net Profit (For 10 years)	2,038,889

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

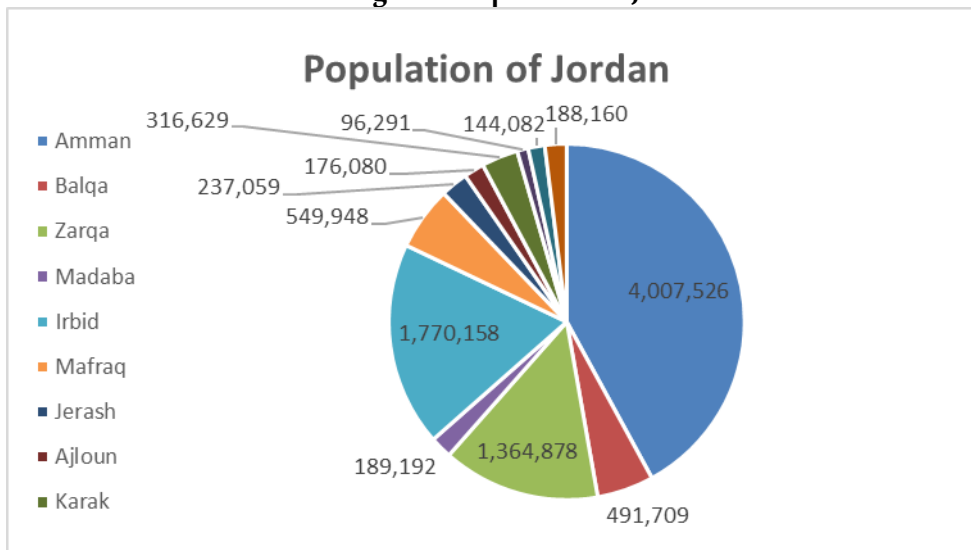
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 1: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

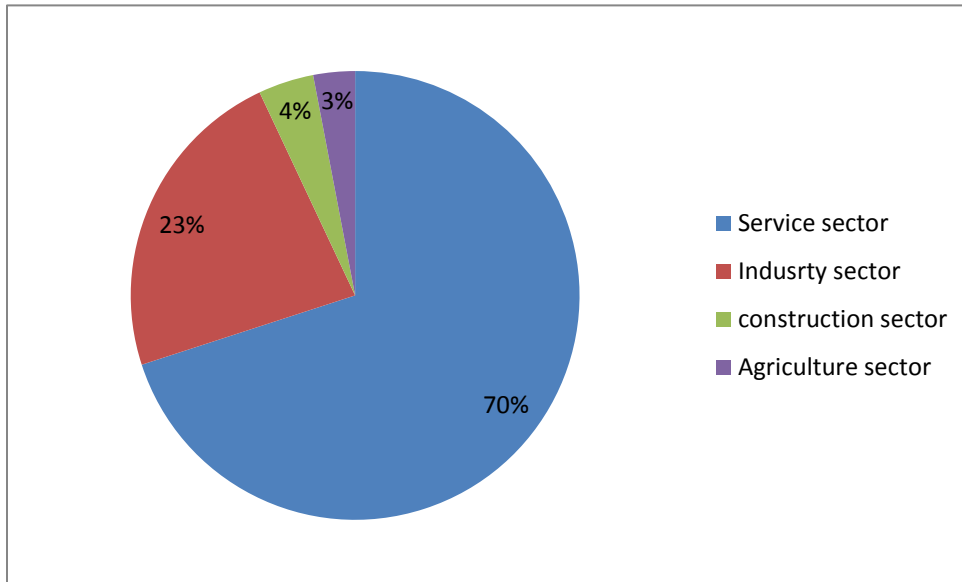
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of

capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Mafrq:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table 2 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20% As per Investment Law categorization, Irbid is classified in Class (C) where investors enjoys 60% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

3. Market Analysis

Project description

Jordan is characterized by a large amount of limestone used in north, middle, east, south of Jordan and the sand stone as well. Names of limestone in Jordan changed from time to time. The stone called by the name of various extracted area or by the physical properties.

The Jordanian limestone has a special properties which is desired for construction purposes such as united colour, texture and clear of pyrite. Low of porosity, voids ratio and very high hardness. For each type of Jordanian limestone a various engineering and physical properties which commercially characterized it than others, these variations related to the geological circumstances participated in its formation.

Types of Jordanian limestone:

- Maan stone: is the best stone in Jordan based on physical properties. Many commercial names are used according to its locations such as Sateh Maan, AlJazeera, Jerdana, and it exist in many grades of quality which classified into; first, second and third class and from the geological point of view the limestone of this type of related to Eocene era.
- Al Rewashed stone: the high-quality types of this type have the same properties of Maan limestones, but it is less whiteness and it was taken from the same layer levels of Maan limestone, this name is according to al Rewashed region in eastern part of Jordan.
- Alehian stone: one of the most common types used because of its accepted properties and reasonable prices, less hardness and more water absorption than previous types, but it is more contrast colour. The name is according to Al Hayan in Mafraq governorate, northeast of Jordan.
- Alsamek stone: It is accepted specifically in the Arabic Gulf states and it has united colour and its boulders allowed cutting it into large sizes. The name is related to the Samek village.
- Qatraneh stone: which is extracted Qatraneh area which located in Karak governorate.

Many properties should be available in order to be used for construction purposes, the Building Research Center at the Royal Scientific Society was conducting many detailed researchs to determine the standard specifications for construction stone submitted by the Department of specifications and standards, and

the Ministry of Public Works, where the stone classified into 3 categories A, B, and C in terms of engineering properties. The table below shows the properties:

Table 2 physical properties of lime stone

The properties	percentage
absorption	(0.5-12) %
Specific gravity	55,47
Cutting force	3.4,5.2,6.9
Bracking resistance	11-123
colour	white
Abrasion resistance	%10
porosity	low

Chemical composition

Table3 chemical properties

component	(%) percentage
CaO	38-42
SiO ₂	20-25
AL ₂ O ₃	2-4
Other oxide like Mg, Na	1.5-2.5
Loss of Ignition	30-32

Project Objectives

- Extract the limestone for construction purposes.
- Increase exporting percents of the Jordanian ores.
- create job opportunities for the local community.

Limestone uses

The limestone one of the most important ores, which enters as main or secondary material in many industrial fields such as:

- Portland cement manufacturing:

Where the limestone represents the basic structure of the industry, where it mixed with non-metallic mineral materials which is chemically treated. It considered the basic material in this industry with a percentage not less than 76%.

- Anhydrous sodium carbonate or soda ash industry

Pure limestone it used for Glass Manufacturing and cleaning materials. The process is carried out by the interaction of carbon dioxide which resulting from the calcination with sodium chloride and ammonia.

- Used as catalyst in melting brass and steel:

The Iron ores usually contains silica, ammonia as impurities which will be removed by the adding the limestone or the lime as main material. Its found that 410kg of limestone is needed to produce 1 ton of Iron.

- Rockwool industry:

The impure limestone is melting in an oven and the slag is skimmed, then a large amount of vibration is produced gives flakes of calcium silicates, after that the flakes are collected as rockwool to be used as thermal isolater.

- Lime manufacturing

Its known industry since centuries which the limestone is burned to lose its carbon dioxide and 44% of its original weight to produce calcium oxide which is a powder iused in many chemical industries. This industry needs tempreature between (1000-1100) °C. The major properties of the lime is high attraction force with water, and gives a high tempreature to produce new types of lime that use in construction industry and as coats the houses also used in glass and paper industry.

- Mineral fillers industry

The limestone is one of the most important materials used as a fillers metal in many sectors, since its easy to milling and less of oil absorption and easy to mix without causing erosion in devices. Also its used in the manufacturing of bitumen rolls used to covers ceilings and walls of the reservoirs to isolate it to avoid leakage. The limestone used as well for the polymer manufacturing such as boats, cars and plastic industries and in plumbing, electrical and furniture industry, the lime stone steel powder is used in the rubber industry. Limestone is also used as well as a stretch material,

Target Segments

- Companies and contracting offices
- Projects management companies
- Quarries

Market Size Analysis

Based on construction development in Jordan which witnesses an increase every year, the next table explain the volum of constructin and numbers of licence and annual building area:

Table 4 Volume of construction in Jordan during the period (2012-2016)

Description	2012	2013	2014	2015	2016
Housing					
Number of licenses	29,940	33,906	36,642	32,882	36,367
Amman	7,956	8,096	8,904	7,596	7,172
Irbid	8,429	10,063	11,118	10,496	11,293
Zarqa	2,804	3,223	3,103	2,765	3,292
Other	10,751	12,524	13,517	12,025	14,610
Size (in thousands)	10,691.2	11,817.6	12,578.4	10,879.0	10,799.7
Amman	6,013.2	6,075.6	6,084.1	4,967.3	4,019.7
Irbid	1,806.0	2,243.1	2,611.9	2,378.1	2,599.3
Zarqa	756.3	942.9	983.4	888.8	1,076.3
Other	2,115.7	2,556.0	2,899.0	2,644.8	3,104.3
For other purposes					
Number of licenses	2,526	2,695	2,936	2,893	3,043
Amman	1,188	1,176	1,334	1,322	1,373
Irbid	431	568	672	657	547
Zarqa	338	423	471	477	600
Other	569	528	459	437	523
Size (in thousands)	2,215.5	2,167.5	2,413.9	2,243.8	2,510.5
Amman	1,604.0	1,338.3	1,616.8	1,457.0	1,618.5
Irbid	131.1	254.8	233.6	268.9	188.6
Zarqa	146.6	175.6	265.2	217.3	294.4
Other	333.8	398.8	298.3	300.6	409.0

Source: monthly bulletin of statistics, the Central Bank, 2016

The total area of construction in the kingdom is about 13,310,000 m², stone building is estimated by 10.2% of full construction space in Jordan. As the size of the Jordanian market of building stone estimated at 1,360,000 m³ per year. The exports about 340,000 m³ of building stone to neighboring countries.

Competitor analysis

There are many competitors in the local markets such as:

- Alrewashed quarries 10 quarries daily production (10-20 m³).
- Alhayan quarries 2 quarries daily production (10-20 m³).
- Building stone imported from west band.

Price analysis and pricing policy

The pricing of building stone for construction works in the local market ranging between 5-15 JD/m², based on data of the local production of ores building stone, there is a great chance to compete with production capacity estimated at 30,000 m³ in addition to opening foreign markets, the table below shows the annual revenues estimated for the project:

Table5 production capacity

Production Capacity	
Size of the extracted raw material m ³	30000
Ratio of Soft stone extracted	24%
Ratio of hard stone extracted	56%
Percentage of Gravel extracted	20%
Conversion factor from cubic meters to square feet	10

Market Share

It was proposed to establish a quarry for the extraction and processing of building stone of about 1.7% market share from the total annual production, which is about (30,000) m³ per year, 100 m³/day based on 300 working days/year.

Estimated Revenues

The estimated revenues of the project for 10 years (2018-2027):

Table 6 estimated revenues of project during the years (2018-2027)

Prices assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price per square meter of soft stone	8	8	8	9	9	9	10	10	10	10
Price per square meter of hard stone	14	14	15	15	16	16	17	17	18	18
Price per square meter of Gravel	5	5	5	5	5	5	5	5	5	5
Soft stone revenues	576000	593280	611078	629411	648293	667742	687774	708407	729660	751549
Hard stone revenues	2352000	2422560	2495237	2570094	2647197	2726613	2808411	2892663	2979443	3068827
Gravel revenues	300000	300000	300000	300000	300000	300000	300000	300000	300000	300000
Total of revenues	3228000	3015840	3106315	3199505	3295490	3394354	3496185	3601071	3709103	3820376

4. Technical Analysis

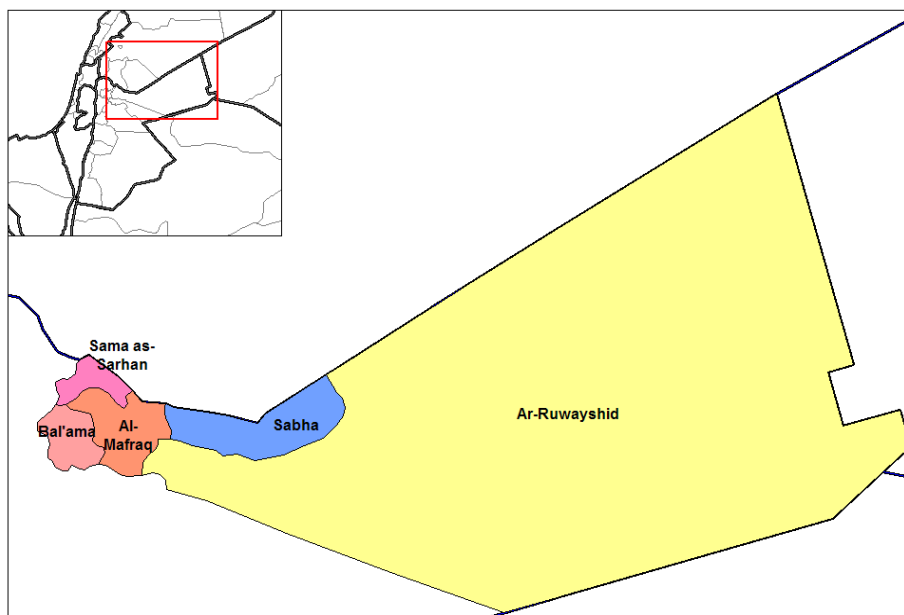
Human resources required

Table7 Human resources required for project

Position	Number
Project manager	1
Technical stone cutter	2
Accountant	1
Labour	2
Driver	2
Security guard	2
Cleaning labour	1
Total	11

Location analysis

Referring to availabilities of building stone occurred Mafrag governate particularly in AlArteen, Tal Rmah areas, the location of quarry will be in these areas:



Description of production process and steps of manufacturing

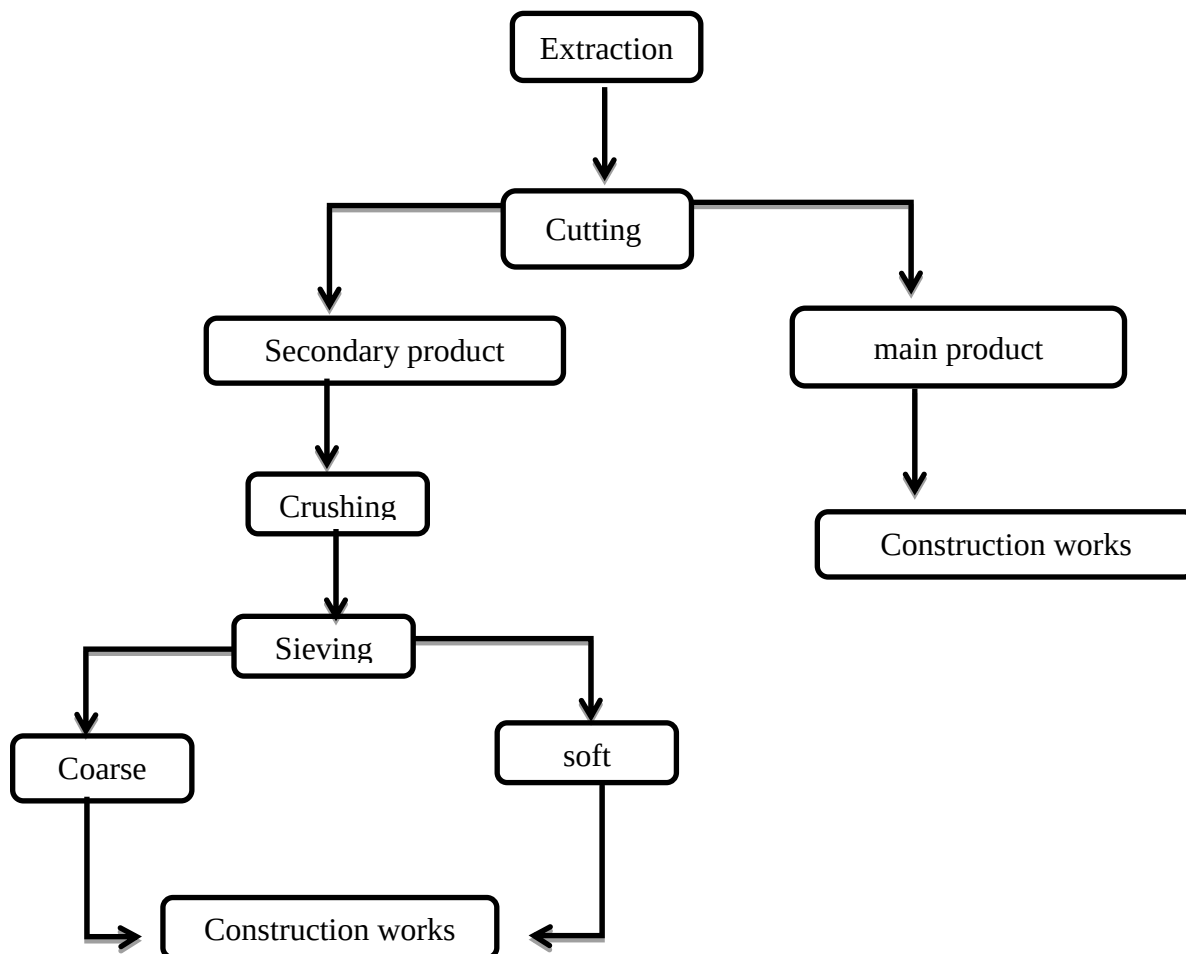


figure 3 Description of processing and manufacturing.

- Extraction the stones from the ground: it's a process of extraction the stone from the rock mountains, and in this stage the usability of stone must be known and ensuring of stone specifications such as hardness, colour and porosity (stone water absorption). large drilling machines used to separate and cut the rocks to large pieces.
- stage of cutting and prepare the stone pieces: after extraction, the stone from the ground the stone will be transported from the quarries to stone saws by large trucks to be cut according to desired measurments and shapes. The cutting devided into several stages, where a disks equipped with diamond pieces on its surruondings, during its rotation to cut the stone, using the technological

development in stone cutting. The equipments and machines where manufactured for cutting the stone by laiser technique as required shapes designd by the computer.

- Crushing stage: The secondary product, will be crushed and sieved it to be used in construction purposes.

Technical requirement analysis

1.1.1 Land and construction works

The table below shows the requirements of construction works:

Table (9): The requirements construction works

Construction works	Area(m ²)
Building works	150
Hangar	500
others	-

Equipments and machinery

The table below shows the equipments and machines for the project:

Table (10): the equipments and machines for the project

Machinery	Qty
Loader	1
Crushing machine	2
Cuttur saw	2
Dumper	1
Sieving machine	1
Dumper	1
Sum.	8

Furnitures

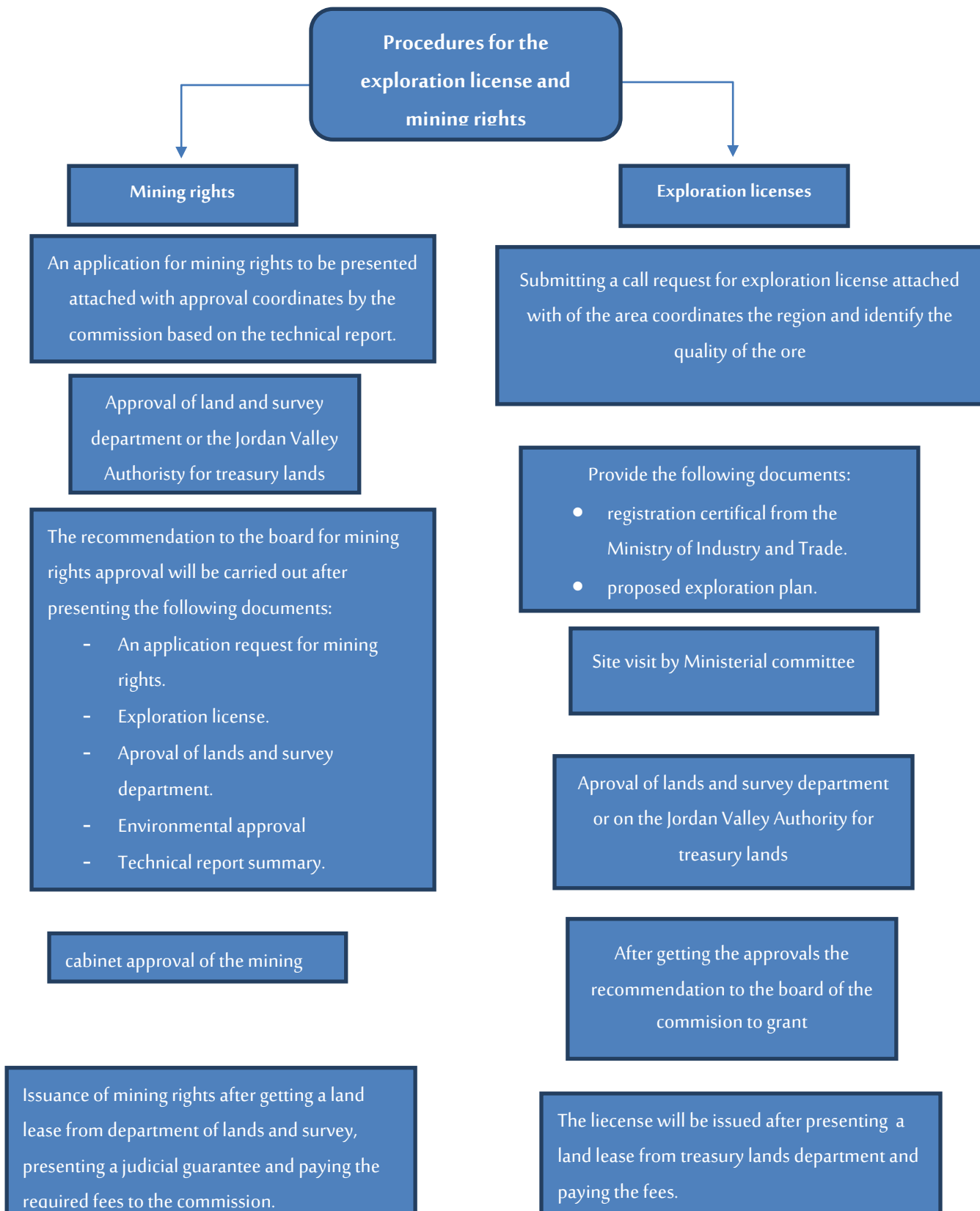
The project is need for these furnitures:

Office furnitures (computers, tables and chaires, carpet, meeting requierments) thet cost 5,000 JD

Requirments and legal procedures

The following figure shows the necessary procedures for exploration and mining rights license depending on body regulating energy and minerals sectors:

figure 2: The legal procedures-Source: Organization of Energy and Mines Commission (TRC).



5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Calculation Assumptions:

- The weighted average cost of capital used in the study is 10.75% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 3 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 4 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 5 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 6 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	65.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	2.5 0%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table 7 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0.50
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	20.00%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table8 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	719,616
Loans Value	1,079,423
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	1,079,423
Loans Percentage	%60

Equity	
Cost of Equity	%15.5
Equity Value	719,616
Equity Percentage	%40
Gross	
Project Value	1,799,039
Weighted Average Cost of Capital	%10.75

Capital Expenditures

The estimated cost of the project is 1,799,039JD and it covers construction works costs, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 9 Construction Work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Construction Works	200	150	30,000
Hunger	100	500	50,000
Other Work			8,000
Total			88,000

Table 10 machinery and equipments Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Loader	1	60,000	60,000
Crushing machine	2	40,000	80,000
Cuttur saw	2	25,000	50,000
Dumper	1	400,000	400,000
Sieving machine	1	25,000	25,000
Dumper	1	200,000	200,000
Total			815,000

Table 11 Furniture and furnishings Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Furniture	-	-	5,000
Total			5,000

Table: 12 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Pick up	2	23,000	46,000
Total			46,000

Table : 13 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table14 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	652,460
General and administrative expenses	3	%25	4,873
Marketing expenses	3	%25	2,018
Indirect Salaries	3	%25	19,640
Total			678,990

Table 15 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Construction works	88,000	10%	96,800	%5.4
Machinery & Equipment	815,000	10%	896,500	%49.8
Furniture	5,000	10%	5,500	%0.3
Vehicles	46,000	10%	50,600	%2.8
Pre-Operating Expenses	2,500	10%	2,750	%0.2
Working capital	678,990	10%	746,889	%41.5
Total (JD)	1,635,490		1,799,039	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,799,039) JD, where 60% of the project will be financed through loans (1,079,423) JD, and the remaining (40%) through (Equity) with (719,616) JD. The following table summarizes the general structure of the required financing:

Table 16 Sources of Funding

Funding Sources	Amount	Percentage
Equity	719,616	%40
Loans	1,079,423	%60
Total	1,799,039	%100
Use of Fund:	Value	Ratio
Capital Expenditures	1,049,400	%58.33
Pre-Operating Expenses	2,750	%0.15
Working Capital	746,889	%41.52
Total	1,799,039	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 17 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from	2,098,20	1,960,29	2,019,10	2,079,67	2,142,06	2,206,33	2,272,52	2,340,69	2,410,91	2,483,24
Utilities from Revenues	161,400	150,792	155,316	159,975	164,774	169,718	174,809	180,054	185,455	191,019
Maintenance from Revenues	80,700	75,396	77,658	79,988	82,387	84,859	87,405	90,027	92,728	95,509
Disposables from Revenues	32,280	30,158	31,063	31,995	32,955	33,944	34,962	36,011	37,091	38,204
Others	237,258	221,664	228,314	235,164	242,218	249,485	256,970	264,679	272,619	280,798
Total	2,609,83	2,438,30	2,511,45	2,586,80	2,664,40	2,744,33	2,826,66	2,911,46	2,998,81	3,088,77

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 18 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	600	618	637	656	675	696	716	738	760	783
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rent	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance	7,871	8,107	8,350	8,600	8,858	9,124	9,398	9,680	9,970	10,269
Miscellaneous	1,772	1,825	1,880	1,936	1,994	2,054	2,116	2,179	2,245	2,312
Total	19,493	20,077	20,680	21,300	21,939	22,597	23,275	23,973	24,693	25,433

Marketing Expenses

Table 19 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	8,070	7,540	7,766	7,999	8,239	8,486	8,740	9,003	9,273	9,551

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table : 20 Staff numbers expected during (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Manager of project	1	1	1	1	1	1	1	1	1	1
Technical cutter stone	2	2	2	2	2	2	2	2	2	2
Accounter	1	1	1	1	1	1	1	1	1	1
Labor	2	2	2	2	2	2	2	2	2	2
Driver	2	2	2	2	2	2	2	2	2	2
Security gurd	2	2	2	2	2	2	2	2	2	2
Cleaning man	1	1	1	1	1	1	1	1	1	1
Total	11	11	11	11	11	11	11	11	11	11

Table 21 Expected monthly salary during (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Manager of project	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Technical cutter stone	380	396	412	429	446	465	484	503	524	546
Accounter	400	416	433	451	470	489	509	530	552	574
Labor	300	312	325	338	352	367	382	397	414	431
Driver	400	416	433	451	470	489	509	530	552	574
Security gurd	300	312	325	338	352	367	382	397	414	431
Cleaning man	250	260	271	282	294	306	318	331	345	359

Table : 22 Expected Annual Salaries and Wges (2018-2027)

المسمى الوظيفي	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Manager of project	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Technical cutter stone	12,160	12,659	13,178	13,718	14,280	14,866	15,475	16,110	16,770	17,458
Accounter	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Labor	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Driver	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Security gurd	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Cleaning man	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Total	78,560	81,781	85,134	88,624	92,258	96,041	99,978	104,077	108,345	112,787

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 23 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	96,800	%5.0	%0.0
Construction Works	896,500	%10.0	%2.0
Machinaries	5,500	%10.0	%5.0
Furniture	50,600	%10.0	%0.0

Table 24 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,049,40	1,067,60	1,086,18	1,105,14	1,124,48	1,144,22	1,164,37	1,184,93	1,205,91	1,227,33
Total Depreciation Expenses	100,100	101,921	103,778	105,674	107,609	109,583	111,598	113,654	115,752	117,893
Total Accumulated Depreciation	100,100	202,021	305,799	411,473	519,081	628,664	740,262	853,915	969,667	1,087,56
Total Additions	0	18,205	18,577	18,958	19,346	19,742	20,147	20,561	20,983	21,414
Total Net Book Values	949,300	865,585	780,384	693,667	605,404	515,564	424,114	331,021	236,252	139,772

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table : 25 Loan Details

Loan Value	1,079,423
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				1,079,423
2019	0	102,329	-102,329	1,181,753
2020	0	112,030	-112,030	1,293,783
2021	0	122,651	-122,651	1,416,434
2022	0	134,278	-134,278	1,550,712
2023	0	147,007	-147,007	1,697,719
2024	0	160,944	-160,944	1,858,663
2025	0	176,201	-176,201	2,034,864
2026	0	192,905	-192,905	2,227,769
2027	0	211,193	-211,193	2,438,962

Income Statement

يبين الجدول التالي قائمة الدخل الشامل للمشروع المقترح:

The following table shows a list of comprehensive income statement for the proposed project:

Table 26 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	3,228,000	3,015,840	3,106,315	3,199,505	3,295,490	3,394,354	3,496,185	3,601,071	3,709,103	3,820,376
Gross Operating Revenues	3,228,000	3,015,840	3,106,315	3,199,505	3,295,490	3,394,354	3,496,185	3,601,071	3,709,103	3,820,376
Operating Expenses	(2,609,838)	(2,438,307)	(2,511,456)	(2,586,800)	(2,664,403)	(2,744,336)	(2,826,666)	(2,911,466)	(2,998,810)	(3,088,774)
Gross Operating Profit	618,162	577,533	594,859	612,705	631,086	650,019	669,519	689,605	710,293	731,602
Gross Profit Percentage	%19	%19	%19	%19	%19	%19	%19	%19	%19	%19
Salaries and Benefits (Indirect Staff)	(78,560)	(81,781)	(85,134)	(88,624)	(92,258)	(96,041)	(99,978)	(104,077)	(108,345)	(112,787)
General and Administraive Expenses	(19,493)	(20,077)	(20,680)	(21,300)	(21,939)	(22,597)	(23,275)	(23,973)	(24,693)	(25,433)
Markeing Expenses	(8,070)	(7,540)	(7,766)	(7,999)	(8,239)	(8,486)	(8,740)	(9,003)	(9,273)	(9,551)
Pre Operating Expenses	(2,750)									
Gross Indirect Expenses	(108,873)	(109,398)	(113,579)	(117,923)	(122,436)	(127,124)	(131,994)	(137,053)	(142,310)	(147,771)
Income before Interest, Depreciation and Tax	509,289	468,135	481,280	494,782	508,650	522,895	537,526	552,552	567,983	583,831
Fixed Assets Depreciations	(100,100)	(101,921)	(103,778)	(105,674)	(107,609)	(109,583)	(111,598)	(113,654)	(115,752)	(117,893)
Income before Tax and Interests	409,189	366,215	377,502	389,108	401,042	413,312	425,928	438,898	452,231	465,938

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Bank Interests	(102,329)	(112,030)	(122,651)	(134,278)	(147,007)	(160,944)	(176,201)	(192,905)	(211,193)	(231,214)
Income Before Tax	306,860	254,185	254,851	254,830	254,034	252,369	249,727	245,993	241,039	234,724
Income Tax	(61,372)	(50,837)	(50,970)	(50,966)	(50,807)	(50,474)	(49,945)	(49,199)	(48,208)	(46,945)
Net Profit	245,488	203,348	203,881	203,864	203,228	201,895	199,781	196,794	192,831	187,779
Net Profit Percentage	%8	%7	%7	%6	%6	%6	%6	%5	%5	%5
Compulsory Reserves	(24,549)	(20,335)	(20,388)	(20,386)	(20,323)	(20,189)	(19,978)	(19,679)	(19,283)	(18,778)
Retained Earnings	220,939	403,952	587,445	770,923	953,827	1,135,533	1,315,336	1,492,451	1,665,999	1,835,000

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 27 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	245,488	203,348	203,881	203,864	203,228	201,895	199,781	196,794	192,831	187,779
Bank Interests	102,329	112,030	122,651	134,278	147,007	160,944	176,201	192,905	211,193	231,214
Depreciation	100,100	101,921	103,778	105,674	107,609	109,583	111,598	113,654	115,752	117,893
Total Operating Cash flows before Additions to Working Capital	447,917	417,299	430,310	443,816	457,844	472,421	487,580	503,353	519,775	536,886
Inventory (Increase/Decrease)	(174,850)	11,492	(4,901)	(5,048)	(5,199)	(5,355)	(5,516)	(5,681)	(5,852)	(6,027)
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Working Capital (Increase/Decrease)	85,273	(5,605)	2,390	2,462	2,536	2,612	2,690	2,771	2,854	2,939
Net Cashflows from Operating Activities	(89,577)	5,887	(2,511)	(2,586)	(2,664)	(2,743)	(2,826)	(2,911)	(2,998)	(3,088)
Cashflows from Investments Activities										
Fixed Assets (Procurement)										
Net Cashflows from Investments Activities	(1,049,400)	(18,205)	(18,577)	(18,958)	(19,346)	(19,742)	(20,147)	(20,561)	(20,983)	(21,414)
Cashflows from Financing Activities										
	(1,049,400)	(18,205)	(18,577)	(18,958)	(19,346)	(19,742)	(20,147)	(20,561)	(20,983)	(21,414)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Capital	719,616									
Loan Amortization	102,329	112,030	122,651	134,278	147,007	160,944	176,201	192,905	211,193	231,214
Bank Interest Rate	(102,329)	(112,030)	(122,651)	(134,278)	(147,007)	(160,944)	(176,201)	(192,905)	(211,193)	(231,214)
Loans	1,079,423	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,799,039	0	0	0	0	0	0	0	0	0
Net (Increase/Decrease) in Cash	1,107,980	404,981	409,222	422,272	435,834	449,936	464,607	479,882	495,795	512,384
Cashflows at the Beginning of Period	0	1,107,980	1,512,961	1,922,182	2,344,455	2,780,289	3,230,224	3,694,832	4,174,713	4,670,508
Cashflows at the End of Period	1,107,980	1,512,961	1,922,182	2,344,455	2,780,289	3,230,224	3,694,832	4,174,713	4,670,508	5,182,892

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 28 Expected Balance Sheet

الوصف	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	1,107,980	1,512,961	1,922,182	2,344,455	2,780,289	3,230,224	3,694,832	4,174,713	4,670,508	5,182,892
Inventory	174,850	163,358	168,259	173,307	178,506	183,861	189,377	195,058	200,910	206,937
Total current Assets	1,282,830	1,676,319	2,090,441	2,517,761	2,958,794	3,414,085	3,884,208	4,369,771	4,871,418	5,389,829
Non Current Assets										
Fixed Assets (net)	949,300	865,585	780,384	693,667	605,404	515,564	424,114	331,021	236,252	139,772
Total Non Current Assets	949,300	865,585	780,384	693,667	605,404	515,564	424,114	331,021	236,252	139,772
Total Assets	2,232,130	2,541,903	2,870,825	3,211,428	3,564,199	3,929,649	4,308,322	4,700,792	5,107,669	5,529,602
Current Liabilities										
Accounts Receivable	85,273	79,668	82,058	84,520	87,056	89,668	92,358	95,128	97,982	100,922
Remaining amount of Loan	(112,030)	(122,651)	(134,278)	(147,007)	(160,944)	(176,201)	(192,905)	(211,193)	(231,214)	0
Total current Liabilities	(26,757)	(42,982)	(52,219)	(62,487)	(73,888)	(86,534)	(100,548)	(116,064)	(133,231)	100,922

Building Stone Project in Mafraq

الوصف	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non Current Liabilities										
Long Terms Loans	1,293,783	1,416,434	1,550,712	1,697,719	1,858,663	2,034,864	2,227,769	2,438,962	2,670,175	2,670,175
Total Long Term Liabilities	1,293,783	1,416,434	1,550,712	1,697,719	1,858,663	2,034,864	2,227,769	2,438,962	2,670,175	2,670,175
Total Liabilities	1,267,026	1,373,451	1,498,492	1,635,232	1,784,775	1,948,330	2,127,222	2,322,897	2,536,944	2,771,097
Owners Equity										
Shareholders Contributions	719,616	719,616	719,616	719,616	719,616	719,616	719,616	719,616	719,616	719,616
Statutory Reseve	24,549	44,884	65,272	85,658	105,981	126,170	146,148	165,828	185,111	203,889
Retained Profits	220,939	403,952	587,445	770,923	953,827	1,135,533	1,315,336	1,492,451	1,665,999	1,835,000
Total Equity	965,104	1,168,452	1,372,332	1,576,196	1,779,424	1,981,319	2,181,100	2,377,895	2,570,726	2,758,505
Total Liabilities and Equity	2,232,130	2,541,903	2,870,825	3,211,428	3,564,199	3,929,649	4,308,322	4,700,792	5,107,669	5,529,602

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 29 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(1,799,039)	361,09	404,98	409,22	422,27	435,83	449,93	464,60	479,88	495,79	512,38	7,890,864
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash	(1,799,039)	265,79	269,16	245,57	228,80	213,22	198,74	185,30	172,81	161,20	150,42	2,316,610

Table below (42) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 30 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(1,799,039)	361,090	404,981	409,222	422,272	435,834	449,936	464,607	479,882	495,795	512,384
Project Value Returned	1,799,039	1,437,949	1,032,968	623,746	201,474	(234,360)	(684,296)	(1,148,903)	(1,628,785)	(2,124,580)	(2,636,964)
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table 31 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	2,608,617
Payback Period	4
Internal Rate of Return	%28.97

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 32 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%28.97	4	10.75%
Revenues declined by 10%	%27.17	4	10.75%
Operating Expenses Increased by 10%	%15.71	8	10.75%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 2,608,617 Jordan Dinars considering that the project provides 11 Job Opportunities for the governorate residents.